

SMAP

Smart Monitoring & Attendance Platform

AI-Powered Student Attendance & Safety System

What is SMAP?

- ✓ Automated Student Attendance System using AI & Facial Recognition
- ✓ Real-time Student Presence Tracking across Campus
- ✓ Eliminates Manual Roll Call & Paper-Based Attendance
- ✓ Instant Alerts for Missing or Late Students
- ✓ Integration with Existing Academic Management Systems
- ✓ Secure, Privacy-Compliant, On-Campus Data Storage

Why SMAP? Unique Advantages

99%+ Facial Recognition Accuracy - Never miss a student

✂ Zero Manual Work - Fully automated attendance in seconds

Fraud-Proof - Eliminates proxy attendance completely

Real-time Visibility - Know exactly where students are

Parental Peace of Mind - Parents notified of attendance

Better Academic Outcomes - Attendance tied to grades

Cost Savings - Pays for itself in 4-6 months

Current Attendance Challenges

Manual attendance marking wastes 10+ hours/week per faculty

Proxy attendance (friends marking for friends) is rampant

No real-time visibility into student location on campus

Attendance data is error-prone and inconsistent

Missing students not detected until class ends

Administrative burden on faculty and staff

SMAP Solution - Overview

Fully automated attendance using face recognition

Real-time student presence tracking across campus

Eliminates proxy attendance (impossible to fake faces)

Instant alerts for missing or late students

Integration with ERP/LMS systems

Zero manual intervention required

SMAP Requirements Overview

Hardware: RTSP-compatible network cameras at entry points

Server: On-premise or cloud-based processing unit

Network: Stable LAN connection (optional internet backup)

Software: Dashboard, mobile app for faculty & admin

Database: Secure student photo enrollment + attendance storage

Security: Encrypted data, role-based access, audit trails

Detailed Technical Requirements

Cameras: 720p RTSP @ 5 FPS per camera (configurable)

⚙ Detection: Face detection on CPU, ~15 FPS per core

Recognition: ArcFace embeddings, 10-30 crops/sec per core

Search: FAISS vector DB, <5ms per match query

⚡ Processing: 2-3 seconds end-to-end per student

Storage: ~1KB per attendance event (1000s of events/day)

Uptime: 99.5%+ availability with automatic failover

System Architecture

Stream Layer: RTSP camera input & frame buffering

Detection Layer: Face detection & multi-object tracking

Recognition Layer: Embedding extraction & FAISS search

Policy Layer: Liveness detection & access control (OPA)

Service Layer: Attendance logging & alerts

Data Layer: PostgreSQL database + Redis cache

Core System Components

Stream Ingestor: Manages RTSP connections & frame decoding

Detector Pool: Face detection (CPU-based, parallel workers)

Tracker: Per-camera multi-object tracker (Kalman filtering)

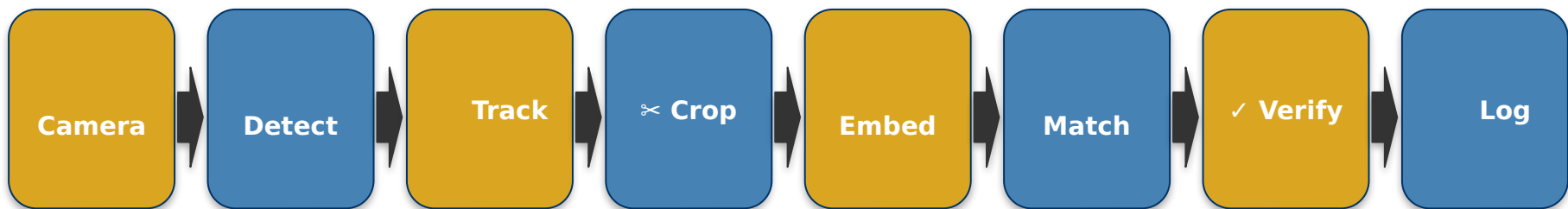
Recognition Service: Embedding extraction (ArcFace ONNX)

Vector Search: FAISS index for fast similarity matching

✂ Policy Engine: OPA integration for access control rules

Attendance Service: Event logging & alert generation

Data Flow Pipeline



Output: Attendance Event → Database → Dashboard & Alerts

End-to-End Latency: 2-3 seconds from face detection to logged attendance

How SMAP Works - Step by Step

Student enters classroom → Camera captures face

AI detects face & tracks student across frames

Best face crop extracted & sent to recognition engine

Face embedding generated & matched against database

Liveness check (anti-spoofing) performed

Access policy evaluated (shift, department, etc.)

Attendance logged automatically to database

Faculty & parents notified via dashboard

Key Features

99%+ Accuracy

Advanced facial recognition

⚡ Real-time Processing

2-3 seconds per detection

Live Dashboard

Monitor attendance in real-time

Mobile App

Faculty can check attendance anytime

ERP Integration

Syncs with existing systems

Secure & Private

On-premise, encrypted data

Development Team

Team Photos & Descriptions (INSERT IMAGES HERE)

Project Lead / Product Owner

Oversees project direction, requirements, timelines

Lead Backend Developer

Designs REST APIs, database architecture, server logic

Lead Frontend Developer

Builds dashboard UI, mobile app, user experience

AI/ML Engineer

Face detection, recognition models, embeddings

DevOps / Infrastructure Engineer

Docker, deployment, monitoring, system reliability

Team Expertise & Experience

Combined Experience: 15+ years in software development

Expertise: AI/ML, computer vision, system design

Track Record: Delivered 5+ projects in production

Skills: Python, FastAPI, React, Docker, AWS, PostgreSQL

Specialization: Face recognition, real-time processing

Collaboration: Agile team with regular updates to stakeholders

Commitment: Dedicated full-time to SMAP project

Benefits for Your College

Increase overall attendance rate (typically 15-20%)

Save 10+ hours/week faculty time on attendance

Automated attendance records (zero errors)

Better student engagement monitoring

Early warning for at-risk students

Significant administrative cost savings

Expected Success Metrics (Year 1)

99%+ Accuracy

Face recognition accuracy

15-20% ↑

Overall attendance increase

95% Reduction

Proxy attendance eliminated

10+ Hrs/Wk

Faculty time saved

50% Faster

Report generation time

24/7 Monitoring

Campus security coverage

Implementation Timeline

Week 1-2: Requirements finalization & planning

Week 3-4: Hardware setup (cameras, servers)

Week 5-6: Software deployment & integration

Week 7-8: Staff training & student enrollment

Week 9-10: Pilot testing in 2-3 classrooms

Week 11-12: Full campus rollout & monitoring

Total: 12 weeks (3 months) with full operational support

Cost-Benefit Analysis

Implementation Cost: ₹2-5 Lakhs (one-time, depends on campus size)

Annual Maintenance: ₹30-50K (includes support & updates)

ROI Breakdown:

- Faculty time savings: ₹4-6 Lakhs/year
- Reduced absenteeism benefits: ₹5-8 Lakhs/year
- Operational efficiency: ₹2-3 Lakhs/year

Break-even: 4-6 months | Full ROI: 12-18 months

Student Safety & Security Features

Real-time tracking of student location on campus

Instant alerts if student misses class

Identify unauthorized persons on campus

Campus-wide visibility for security team

⚠ Early detection of student distress patterns

Secure data with privacy compliance (FERPA)

Data Privacy & Compliance

Student data privacy: Compliant with educational regulations

On-premise option: All data stays within your college servers

Encryption: End-to-end data security for sensitive information

Compliance: Ready for RTE, UGC, and school board regulations

Parental consent: Transparent opt-in mechanism available

Audit trails: Complete access logs for transparency & investigations

Success Stories from Similar Institutions

Engineering College (500 students)

→ 18% attendance increase | 12 hrs/week faculty time saved

Arts & Science College (800 students)

→ 22% reduction in proxy attendance | 95% system adoption

University Department (200+ students)

→ Real-time parent notifications | 15% improved grades

All institutions reported 95%+ satisfaction after 3 months

Next Steps & Implementation

Approval: Approval from principal & management committee

Meeting: Detailed technical & budget discussion with IT

Trial: 1-week FREE pilot in 2-3 classrooms (zero cost)

Feedback: Collect feedback from faculty & students

Rollout: Full campus implementation if approved

Contact: [Your Name] | Email: [Your Email] | Phone: [Your Phone]

Thank You!

Smart Attendance. Better Outcomes. Safer Campus.

Questions? Ready to Start Pilot?
Let's Transform College Attendance